RUBRA: An Agentic AI System for Automatic Short Answer Grading Using LLMs and RAG

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***Abstract*—The proliferation of deep Learning applications in natural language processing has facilitated automated evaluation of short-answer questions, providing more transparent, interpretable and unbiased evaluation of students' responses. In this work, we introduce RUBRA, an agentic AI system for Automatic Short Answer Grading (ASAG) using Large Language Models (LLM) and Retrieval-Augmented Generation (RAG) techniques. The system incorporates domain-specific context information retrieved from textbooks and study materials through a RAG framework and a dynamically generated rubric from reference answers provided by human tutors. Then, by aligning student responses with the rubric via semantic similarity scoring, the system computes an explainable and fair grade and provides transparent feedback to the learner. We discuss the architecture, components, and potential effectiveness of the proposed agentic AI system and tested our results over the SCIENTSBANK dataset. It is observed that the proposed approach not only outperformed the existing approaches in prediction accuracy but also significantly improved the autonomy and adaptability of the system.**

***Index Terms*—Agentic AI, LLM, RAG, NLP, ASAG**

I. Introduction

Automated evaluation of open-ended questions remains a challenging task due to its variability in correct responses. Conventional ASAG systems aim to address these challenges by providing text-based similarity between students' answers and a reference answer commonly supplied by the tutor [1], [2]. Recent ASAG approaches engage rubric-based assessments to avoid inconsistencies and biases. A rubric is a structured evaluation guide that outlines the key components, concepts, and criteria for grading responses, traditionally used to evaluate long-form or essay-type answers [3]. Although rubrics are primarily associated with extended responses, they are equally valuable in grading short responses, as they bring clarity, fairness, and consistency to the evaluation process. In short answer grading, a rubric helps define what constitutes a complete, partially correct, or incorrect response by identifying essential concepts and acceptable variations in expression. However, manually constructing a rubric for each individual question by identifying key ideas, weighting their importance, and mapping acceptable alternative phrasings, is a tedious and cognitively demanding task, especially across diverse subjects and large-scale assessments. On the flip side, automatic rubric generation for short answer can suffer from lack of conciseness in selecting the dimensions. If generic dimensions like recall, cognition, analysis abilities are chosen then model's performance can have high variance whereas for more specific dimensions like definition, examples, and applications can cause high biases. In this regard, an AI agent, powered by a Large Language Model (LLM), can play a transformative role [4]. By analyzing reference answers, domain materials, and learning objectives, the agentic system can automatically generate detailed, context-sensitive rubrics in a seamless and scalable way.

This research proposes an agentic AI system for ASAG systems leveraging LLM with Retrieval-Augmented Generation (RAG). In conventional approach, different task-specific ML models can be employed for distinct components such as rubric generation, information retrieval, answer evaluation, and feedback generation. However, each of these models requires separate independent training and fine-tuning, which makes the integration and coordination more complex and fragile. In recent times, agentic AI approach has emerged as a competent alternative to this focused and task specific approach [5] [6]. An agentic system can delegate each task to a specialized autonomous agent, such as a Rubric Generation Agent, a Retrieval Agent (RAG), and an Answer Evaluation Agent, all orchestrated within a unified LLM-powered framework. These agents can dynamically communicate, share context, and adapt their behavior based on the specific grading scenario [7] [8].

The objectives of the proposed RUBRA system are the following core tasks: generate rubrics by extracting

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key components from a reference answer using LLM, incorporating domain context by retrieving information from subject-related textbooks and resources, compare student responses against rubrics to assign scores based on semantic similarity, and provide meaningful feedback.

II. Related Works

Automated evaluation of short answers has gained significant attention due to the increasing demand for scalable and objective assessment methods. Traditional grading methods are tedious and prone to inconsistencies, motivating the exploration of LLMs for automated evaluation. Several recent studies have proposed innovative frameworks to improve the accuracy, fairness, and interpretability of automated grading systems by integrating advanced language models and retrieval-based techniques.

Xie et al. [9] introduced the Grade-Like-a-Human framework, a multi-agent LLM-based grading system to addresses the grading process. It includes rubric generation, grading, and post-grading review. This system mirrors human evaluators by refining rubrics based on actual student responses and conducting post-grading checks to enhance fairness and accuracy. The major contribution of this work is the iterative rubric generation using self-reflective prompts. A systematic sampling strategy to capture the diversity of student responses is also proposed. This approach demonstrates improved grading reliability both on a curetted dataset and the widely used Mohler dataset. In a similar work, Hashemi et al. [10] proposed the LLM-RUBRIC framework that focuses on multi-dimensional evaluation through fine-tuned LLMs. This framework uses a manually constructed rubric to evaluate responses across multiple criteria such as conciseness, naturalness, and citation quality. A personalized calibration network is introduced to align LLM predictions with human judgments. Notably, the proposed system achieves a big improvement in prediction accuracy compared to uncalibrated LLMs in the evaluation of dialogue systems.

Farzi et al. [11] introduced a novel approach by combining RAG with automated rubric-based evaluation. This system decomposes an answer into detailed information needs to be addressed. The evaluation process is conducted in three phases: designing grading rubrics, automatically grading system responses, and generating final evaluation scores. This hybrid model encompasses human defined rubric criteria, while LLMs handle the actual grading. This improves both efficiency and reliability of the automated assessments. Moreover, this study addresses concerns about the reliability of LLM-based grading by incorporating a human-in-the-loop model to refine rubrics iteratively. Sengupta et al. [12] proposed a fusion model for the ASAG task, which combines the advantages of GNN-based textual features providing document-level structural information with the semantic and contextual embedding of the transformer-based BERT model. An empirical study on the publicly accessible ASAP dataset achieved a 0.885 F1 score in the four-class classification task, outperforming the other state-of-the-art techniques.

Comparing these literatures reveals multiple scopes of work in ASAG. First, rubric generation remains a critical challenge in ASAG systems. Second, finetuning LLM models pose challenges due to inherent model variability [13]. Third, the integration of RAG for improving rubric's completeness is promising but yet to be established [14]. Fourth, very few ASAG sytems tried to provide reliable feedback on the assessment to make the process more transparent [15], [16]. Finally, although Agentic AI systems have recently gained much attention across various domains, their application to the ASAG task remains largely unexplored, despite significant potential for impact. Our proposed work integrates multiple models like LLMs, RAG, and semantic similarity models, for rubric-based assessment of short answers using Agentic AI.

III. Methodology

This work presents RUBRA, a robust and explainable ASAG system leveraging an agentic AI architecture where each functional task is delegated to a dedicated autonomous agent. These agents include: (i) the rubric generation agent, (ii) the contextual retrieval agent, (iii) the answer evaluation agent, (iv) the feedback generation agent, and (v) the orchestration agent. Each agent is powered by separate LLMs and operates in coordination to ensure consistency, contextual awareness, and adaptability. Fig. 1 depicts the agentic AI architecture for the proposed work and the rest of the section details the internal workings of each agent and their role in the grading pipeline.

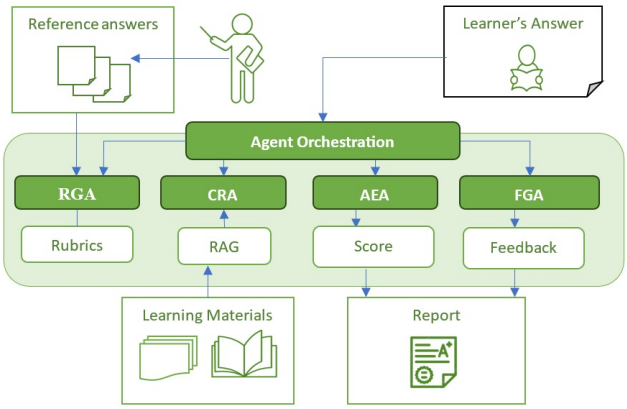

Fig. 1. The Framework of the proposed RUBRA model

A. Rubric Generation Agent (RGA)

The Rubric Generation Agent is responsible for dynamically constructing rubrics based on provided sample or reference answers. These reference responses are typically

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collected from subject-matter experts such as teachers, archived grading records, or online educational resources. RGA uses a pretrained LLM for text embeddings using a sentence-transformers model called all-MiniLM-L6-v2. It synthesizes commonalities across the input answers and decompose them into structured rubric elements. Each rubric entry includes elements like the key concept to be evaluated, key words with acceptable alternative phrasings, formulas, equations (if any), importance weighting, and error tolerance. Identifying these elements can be achieved by fine-tuning or prompting the LLM using few-shot examples. The model employs Part of Speech (POS) tagging, Named Entity Recognition (NER) and semantic clustering via Sentence-BERT embeddings to identify semantically similar concepts across answers. The output is converted into a structured JSON format for seamless communication across multiple agents. The final rubric containing all identified values with possible variations with individual weights, serves as reference points for answer evaluation.

B. Contextual Retrieval Agent (CRA)

The CRA is designed to augment rubric content and evaluation logic by retrieving external knowledge from domain-specific materials such as textbooks, lecture notes, or curated web content. This is critical for capturing edge cases where student answers include valid, yet unanticipated, phrasing or supplementary concepts that are not present in the original reference answers. The retrieval pipeline is built using a hybrid RAG framework combining dense vector retrieval with keyword-based filtering. Initially, all supplied study materials are chunked into context windows of 200–300 tokens and embedded using the sentence-level transformer *all-MiniLM-L6-v2*. These embeddings are stored in a vector database using *FAISS*, indexed by topic or subject hierarchy. Then the rubric is updated by the RAG agent by invoking with a prompt. A query embedding is computed and matched against the top-K closest passages using cosine similarity. The retrieved documents are then used to refine the rubric by expanding acceptable variations. Similarly, when the evaluation agent encounters low-confidence matches, the RAG agent is invoked to resolve ambiguous grading scenarios. In addition, metadata like textbook name, chapter, page number is retained and forwarded to the feedback agent to enhance transparency of the evaluation. The RAG component ensures that the system can generalize beyond rigid answer templates and embrace open-ended student creativity within pedagogically valid limits. This ensures the adaptability of the system.

C. Answer Evaluation Agent (AEA)

The AEA evaluates a student's answer by comparing it with the structured rubric and calculating a numeric score based on semantic alignment. First, the student's answer is embedded using the same transformer encoder as used in each rubric component. For each rubric entry r_i, the cosine similarity with the student response s is computed as $sim(r_i, s)$. Formally, let $R = r_1, r_2, ..., r_n$ be the rubric concepts with associated weights $W = w_1, w_2, ..., w_n$ and let r_i and s be their embeddings. Then the similarity score S is computed as eq. 1, where σ is a soft-thresholding function to penalize low similarity matches. This allows partial correctness of answer based on the strength of semantic overlap. The final score (FS) is derived by normalizing the weighted sum against the total rubric weight as shown in eq. 2.

$$S = \sum_{i=1}^{n} w_i \cdot \sigma(cosine(r_i, s)) \tag{1}$$

$$FS = \frac{S}{(\sum_{i=1}^{n} w_i)} \times 100 \tag{2}$$

The AEA agent also has the provision to flags uncertain or borderline matches for optional human-in-the-loop (HITL) review or go back to the CRA agent for further refinement of the rubric. The evaluation logic is designed to be interpretable, and matching pairs are shared for further processing by the feedback agent to ensure fairness and traceability.

D. Feedback Generation Agent (FGA)

The FGA enhances the pedagogical value of the system by providing detailed, actionable, and personalized feedback to students. It focuses on identifying rubric items that were partially matched, or entirely missing from the student answer. These gaps are extracted by the AEA and passed to the FGA. The feedback is enhanced by referencing the corresponding text materials retrieved earlier by the RAG agent. To incorporate personalized feedback and avoid generic suggestions, the agent leverages few-shot examples contextualized by the subject area and student proficiency level. In addition, common student misconceptions are stored in a lookup table, allowing the agent to tailor explanations if the student exhibits typical error patterns. The generated feedback is formatted into a summary card that includes matched concepts, missing elements, and improvement tips. This helps learners understand not just their score, but also the reasoning of the evaluation process, making the system more than a grader, but a tutor as well.

E. Orchestration and Synchronization Agent (OSA)

The OSA lies at the core of the system to supervise task allocation, state management, and inter-agent communications. The architecture follows a modular pipeline pattern of rubric construction $\rightarrow$ retrieval $\rightarrow$ evaluation $\rightarrow$ feedback generation. It uses *FastAPI* to ensure a smooth communication with the environment and loose

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coupling between agents. The OSA works as an event-driven controller orchestrating the sequence. Each agent registers with the orchestrator and subscribes to task events via an internal publish-subscribe model. The context information is passed using shared memory structures of serialized JSON payloads, with explicit keys for rubric items, embeddings, and intermediate scores. When the AEA flags uncertain cases the orchestrator dynamically decides whether to trigger the agent RAG. The orchestration module also supports domain adaptation by making the agent logic largely reusable. For introducing any new subject, only the knowledge base and prompt templates need to be updated. This ensures extensibility of the system by making it viable across multiple educational levels and disciplines.

IV. Results and Discussion

We present an empirical validation of our proposed system for ASAG task using the widely-used and publicly available SCIENTSBANK dataset. An experimental study is performed to compare the performance of RUBRA to existing ASAG baselines.

A. Dataset

The SCIENTSBANK dataset is part of the SemEval-2013 shared task on student response analysis. It comprises short answers to open-ended science questions which are annotated by human graders. It contains 10,804 answers to 197 assessment questions across 15 different science domains. The answer length varies between 5 to 50 words. Each response is labeled into five categories such as correct, partially correct or incomplete, contradictory, irrelevant, and non-domain.

B. Experimental Setup

The proposed system was implemented using five collaborating agents with synchronized interactions. Each of the module is allowed for independent upgrades and task delegation. For a comprehensive performance comparison, we evaluated it against three strong baseline models : BERT + Linear Classifier (BERT-LC), BERT + Semantic Similarity + Linear Classifier (BERT-SSL), and GPT-3.5 Grader zero-shot (GPT3.5-ZS) on the SCIENTSBANK dataset. The BERT-LC baseline fine-tunes a pretrained BERT model- *bert-base-uncased* (110M parameters). The CLS token embedding is passed through a fully connected classification head to predict one of the five classes. The training used cross-entropy loss, Adam optimizer, learning rate 2e-5, and batch size 16. The second baseline, BERT-SSL, uses sentence embeddings generated by *Sentence-BERT* (SBERT) to measure cosine similarity between student answers and the reference answer. These normalized similarity scores are then used as augmented features to predict the final class. The third baseline, GPT-3.5-ZS model is implemented by OpenAI *GPT-3.5-turbo* model with a handcrafted prompt that includes the question and the reference answer, and the student response. The model is asked to assign a grade directly in zero-shot fashion using GPT-3.5's generative reasoning abilities.

C. Result Analysis

Table I presents the performance comparison of the proposed Agentic AI approach against three baselines. It can be observed that RUBRA outperformed the baselines by a considerable margin in all performance metrics. Table II details the class-wise performance of the models. It is observed that the RUBRA exhibits significant improvements in *Partially Correct* and *Contradictory* categories, which are otherwise found to be difficult to handle by other classification models. This improvement is attributed to rubric decomposition and RAG-assisted semantic matching.

TABLE I
Comparison of performance of RUBRA with baseline models

Method	Accuracy (%)	Macro-F1 (%)	Precision (%)	Recall (%)
BERT-LC	61.2	58.7	60.1	57.4
BERT-SSL	64.3	60.9	63.5	59.8
GPT3.5-ZS	69.2	65.1	69.5	64.8
RUBRA (Proposed)	71.5	69.8	70.3	68.4

TABLE II
Class-wise performance of the models

Class	BERT-LC	BERT-SSL	GPT3.5-ZS	RUBRA
Correct	77.2	78.5	82.35	86.1
Partially Correct	52.5	55.4	59.48	68.9
Contradictory	43.8	49.2	57.51	62.3
Irrelevant	55.6	59.1	64.28	67.4
Non-domain	64.0	66.5	70.54	71.2

The bar chart in Fig. 2 illustrates the variation in model accuracy across different answer length categories, offering insights into how each grading method handles increasing complexity in student responses. While BERT-LC performed weakly in all bin segments and showed less stability across varying lengths, BERT-SSL maintained steady improvements with increasing length, benefiting from semantic alignment. GPT3.5-ZS showed a competitive performance for mid to long-length answers but less effective in shorter answers, indicating limitations in zero-shot reasoning without tailored prompts. The proposed RUBRA model consistently outperformed all baselines across all length bins, with its accuracy improving from 66.5% for very short answers (less than 10 words) to 75.2% for longer responses (more than 40 words). This represents a strong generalization and contextual understanding by the model. The performance across different answer lengths highlights the system's robustness. This improvement is due to better identification of key elements within the answer via the rubric and RAG components. The confusion matrix in Fig. 3 highlights key model behaviors. The *Correct* class has high true positive values

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(86. 1%), confirming the reliability of the model in grading well-structured responses. However, *Contradictory* and *Irrelevant* classes exhibit notable confusion, often misclassified as *Partially Correct*. This can be caused from superficial keyword overlaps despite semantic contradiction. However, despite some failure cases, the overall performance highlights the efficiency of the proposed rubric-guided, context-aware agentic system in handling both concise and elaborate responses effectively.

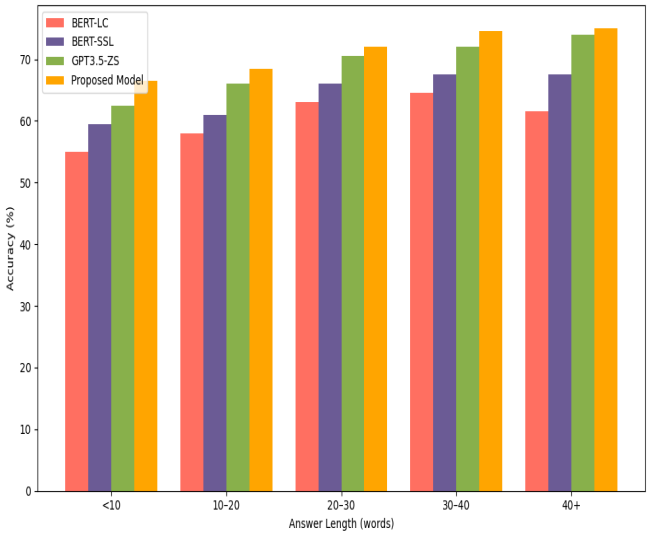

Fig. 2. Performance accuracy of the proposed RUBRA model at different length of answers

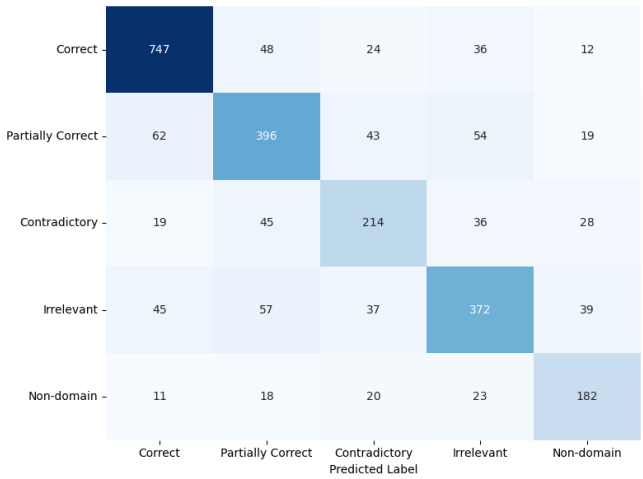

Fig. 3. Confusion matrix of the performance of the RUBRA model

D. Ablation Study on Agent Contributions

To assess the significance of each agent in the RUBRA system pipeline, we conducted an ablation study by systematically disabling one agent at a time and measuring the resultant effect on grading performance. This empirical study helps to quantify the contribution of each module and validate the necessity of the proposed agentic architecture. Table III presents the results of the ablation study.

The removal of the context retrieval agent caused a noticeable drop of 7.4% in accuracy, highlighting its role in enriching the rubric with additional domain-relevant context to handle uncommon or paraphrased student inputs. The absence of the rubric generation agent led to a larger performance degradation, indicating the central role of structured rubrics in guiding semantic alignment. Disabling the answer evaluation agent and replacing with one-shot prediction resulted in the most significant drop in performance, demonstrating its high importance in precise scoring based on rubric alignment. The feedback agent, although pedagogically important, showed minimal impact with 1.5% drop in the scoring accuracy. In general, removing any agentic component increased instability, emphasizing their role in the system.

TABLE III
ABLATION STUDY: PERFORMANCE WITH EACH AGENT REMOVED

Agent Removed	Accuracy (%)	Macro-F1 (%)	Impact
None	71.5	69.8	Baseline with Full RUBRA
RGA	64.2	61.5	Weak rubric generation
CRA	66.3	63.1	Poor contextual adaptability
AEA	60.1	56.9	Inaccurate semantic scoring
FGA	70.4	67.7	No feedback generation

These findings empirically validate the contribution of each autonomous agent in achieving robust, interpretable, and pedagogically meaningful grading of short-answers within the proposed framework.

V. CONCLUSION

This paper presents RUBRA, an Agentic AI-based system for Automatic Short Answer Grading (ASAG) that integrates multiple specialized agents to deliver accurate, interpretable, and scalable evaluation of student response. Unlike traditional task-specific models, our approach leverages the modularity and autonomy of agents to dynamically generate detailed rubrics using LLMs, retrieve domain-relevant context using RAG, evaluate student answers through rubric-guided comparison, and give feedback in a pedagogically meaningful way. An experimental study on the SCIENTSBANK dataset demonstrated significant improvements over baseline models. RUBRA effectively handled diverse answer structures and varying lengths by integrating rubric-guided, context-aware agents. An ablation study of the components established their significance within the framework. Overall, this work confirms that integrating LLMs with structured agentic workflows leads to more accurate and explainable ASAG outcomes, opening new pathways for intelligent systems

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and automated assessment platforms. In the future, the proposed architecture can be further extended to incorporate multimodal inputs like diagrams and equations to adapt to various academic domains.

REFERENCES

[1] G. Verma, S. Sarkar, D. Pillai, H. Shiokawa, H. Shahbazi, F. Veazey, P. Hubbert, H. Su, and P. Buitelaar, "Hybridcontextqa: A hybrid approach for complex question answering using knowledge graph construction and context retrieval with llms," 2024.

[2] S. Haldar, S. Sengupta, and A. K. Das, "A comparative assessment of the contemporary models for automatic short answer grading," in *Computational Intelligence in Pattern Recognition*, (Singapore), pp. 583–594, Springer Nature Singapore, 2025.

[3] S. Datta and S. Sengupta, "A review on the adaptive features of e-learning," *International Journal of Learning and Teaching*, no. 4, pp. 277–284, 2018.

[4] E. Latif and X. Zhai, "Fine-tuning chatgpt for automatic scoring," *Computers and Education: Artificial Intelligence*, vol. 6, p. 100210, 2024.

[5] D. B. Acharya, K. Kuppan, and B. Divya, "Agentic ai: Autonomous intelligence for complex goals–a comprehensive survey," *IEEE Access*, 2025.

[6] S. Hosseini and H. Seilani, "The role of agentic ai in shaping a smart future: A systematic review," *Array*, p. 100399, 2025.

[7] L.-H. Chang and F. Ginter, "Automatic short answer grading for finnish with chatgpt," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, pp. 23173–23181, 2024.

[8] L. Hughes, Y. K. Dwivedi, T. Malik, M. Shawosh, M. A. Albashrawi, I. Jeon, V. Dutot, M. Appanderanda, T. Crick, R. De', *et al.*, "Ai agents and agentic systems: A multi-expert analysis," *Journal of Computer Information Systems*, pp. 1–29, 2025.

[9] W. Xie, J. Niu, C. J. Xue, and N. Guan, "Grade like a human: Rethinking automated assessment with large language models," *arXiv preprint arXiv:2405.19694*, 2024.

[10] H. Hashemi, J. Eisner, C. Rosset, B. Van Durme, and C. Kedzie, "Llm-rubric: A multidimensional, calibrated approach to automated evaluation of natural language texts," *arXiv preprint arXiv:2501.00274*, 2024.

[11] N. Farzi and L. Dietz, "Pencils down! automatic rubric-based evaluation of retrieve/generate systems," in *Proceedings of the 2024 ACM SIGIR International Conference on Theory of Information Retrieval*, pp. 175–184, 2024.

[12] S. Sengupta, M. Mandal, and A. K. Das, "A bert and text graph neural network-based fusion model for automated short answer grading," in *International Conference on Emerging Applications of Information Technology*, pp. 155–164, Springer, 2024.

[13] C. Impey, M. Wenger, N. Garuda, S. Golchin, and S. Stamer, "Using large language models for automated grading of student writing about science," *International Journal of Artificial Intelligence in Education*, pp. 1–35, 2025.

[14] M. Fateen, B. Wang, and T. Mine, "Beyond scores: A modular rag-based system for automatic short answer scoring with feedback," *IEEE Access*, 2024.

[15] I. Estévez-Ayres, P. Callejo, M. Á. Hombrados-Herrera, C. Alario-Hoyos, and C. Delgado Kloos, "Evaluation of llm tools for feedback generation in a course on concurrent programming," *International Journal of Artificial Intelligence in Education*, pp. 1–17, 2024.

[16] A. Filighera, J. Tschesche, T. Steuer, T. Tregel, and L. Wernet, "Towards generating counterfactual examples as automatic short answer feedback," in *International Conference on Artificial Intelligence in Education*, pp. 206–217, Springer, 2022.

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